American Computer Science League

2022-2023 ● Contest 1: Short Problems ● Senior Division

1. Computer Number Systems

In the sequence of decimal numbers from 100 to 129, inclusive, each converted to octal, how many total times do the digits 5 through 7 occur?

A. 36
B. 12
C. 24
D. 48
E. 30

2. Computer Number Systems

What is the value of $FA8072_{16} - 40E0D0_{16}$ in base 8?

A. 56310642
B. 56417642
C. 56317642
D. 56217642
E. 46317642

3. Recursive Functions

Find $f(17)$ given:

$$f(x) = \begin{cases} f(x-5)+x & \text{if } x > 4 \\ f(x+2)-x & \text{if } 2 \le x \le 4 \\ -x & \text{if } x < 2 \end{cases}$$

A. 5
B. 18
C. 6
D. 35
E. -1

4. Recursive Functions

Find $f(12, 18)$ given:

$$f(x,y) = \begin{cases} f(x/2, y-3) - xy & \text{if } x \text{ is even} \\ f\left(y+2, \frac{x}{3}\right) + x & \text{if } x \text{ is an odd multiple of } 3 \\ xy & \text{otherwise} \end{cases}$$

A. -115
B. -331
C. -25
D. -216
E. -14

5. What Does This Program Do?

What is the output of the following program if n = 20?

```
input n
t = 0
for x = 3 to n step 3
        d = 2
        while d < x
             t = t + 1
             d = d * 2
        end while
next x
output t
```

A. 12
B. 13
C. 14
D. 15
E. 16